Semantic Epistemic Risk versus Confidence-Based Active Retrieval During Generation

Working Draft

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Abstract

Active retrieval during generation is established prior art: FLARE retrieves when forecast tokens have low model confidence [1]. We ask whether uncertainty is equivalent to epistemic risk. An initial 80-example controlled freeze finds that a transparent semantic trigger has 100% retrieval recall and 2.5% false activation across Qwen3-0.6B, SmolLM2-1.7B, and Gemma3-1B, while confidence is checkpoint-dependent. We then freeze a 192-example natural-language extension after rejecting four candidate freezes for lexical triviality, duplicates, source-key collisions, or inconsistent gold answers. On its 64-example test, the original shallow lexical accuracy is at most 85.94%. A matched tokenizer-aware audit changes that boundary: native Qwen, SmolLM2, and Gemma token-bigram TF–IDF classifiers each reach 100%, while a shared word/punctuation tokenizer reaches 87.5%. The unchanged semantic policy reaches 100% recall and 12.5% false activation; confidence recall is 96.88%, 81.25%, and 78.12% with false activation of 62.5%, 0%, and 9.38%. Advisory evidence leaves UCR of 18.75%, 12.5%, and 3.12%; an explicit support contract leaves 12.5%, 12.5%, and 0%. Deterministic runtime enforcement reaches zero UCR and full authorized coverage for every family. Prior rank-8 LoRAs have zero held-out recall, and the consumed natural freeze is not reused for adaptation. The result supports transparent runtime enforcement, but not superiority over stronger model-token lexical triggers. The tokenizer result exposes residual template structure and requires a harder independently frozen benchmark.

1 Question and Claim Boundary

The primary question is: *is epistemic risk distinct from model uncertainty?* Retrieval-required means that a reliable answer depends on an external or source-specific record rather than model weights or already supplied active context. This operational label is distinct from model confidence.

The secondary question is placement: *where should stable one-source interface knowledge live: context, weights, or runtime?* Phase B begins only after the Phase A source, benchmark, confidence probes, threshold objective, and evaluation are frozen. The paper does not claim active retrieval itself as novel and does not study multi-source routing.

2 Mechanism

$$Q \to y_{1:t} \to \widehat{y}_{t+1:t+k} \to \{T_{\mathrm{conf}}, T_{\mathrm{sem}}\} \to \text{retrieve} \to \text{evidence/status} \to \text{accept or abstain.}$$

The FLARE-like controller greedily forecasts one short answer span, records the probability of every forecast token, triggers on the minimum probability, and regenerates with evidence. Canonical FLARE forecasts longer text, may mask uncertain spans, and iterates during long-form generation [1]; our single-opportunity controlled diagnostic is not a canonical reproduction.

The semantic controller uses answer-independent lexical, entity, temporal, freshness, source-dependence, and active-context rules. It has an explicit exception for "electric current" and treats compute requests as belonging to a different coprocessor. Rules never contain source answer values.

3 Controlled Source and Evidence Policy

The frozen source is `atlas-controlled-registry@2026.08.2-frozen-candidates`. Records have stable IDs, validity intervals, source version, and provenance. They cover stable, superseded familiar, unfamiliar private, unavailable, and conflicting facts. Requests contain entity, attribute, as-of date, forecast, source version, and resource bounds. Outcomes include SUPPORTED, CONTRADICTED, UNVERIFIED, STALE, AMBIGUOUS, and CONFLICT.

The runtime gate enforces support relative to this configured source; it does not establish truth. A unique supported value may be accepted. Missing, ambiguous, or conflicting evidence forces abstention rather than an unsupported factual commitment.

4 Frozen Measured-Quadrant Benchmark

We generate 256 candidates: 32 development and 224 candidate-test examples. Confidence is measured on pinned Qwen3-0.6B before test selection. The development threshold maximizes retrieval-requirement F1 minus 0.25 times retrieval rate and equals 0.84993. The candidate pool contains 25, 87, 22, and 90 examples in the four observed Qwen quadrants; the freezer selects exactly 20 from each. IDs are fixed before any condition evaluation.

Retrieval-required subclasses are fresh/current, source-version-specific, private, changed familiar, unavailable, conflicting, exact attribution, and structured one-source values. Retrieval-not-required subclasses are stable facts, supplied context, freshness distractors, quotations, hypotheticals, non-factual prose, and compute needs. The final benchmark contains 32 development and 80 untouched test examples.

Each model independently fits its threshold on the common development set: 0.84993 for Qwen, 0.14693 for SmolLM2, and 0.99971 for Gemma. Qwen and Gemma observe all four test quadrants. SmolLM2 places no test example below its fitted threshold, so its test set lacks low-confidence quadrants. This is a calibration result and limits cross-model quadrant claims.

5 Conditions and Metrics

Phase A has 13 conditions: LLM only, anti-hallucination prompting, upfront RAG, explicit retrieval, FLARE-like confidence, semantic, confidence OR semantic, confidence AND semantic, retrospective verification, evidence advisory, evidence plus abstention instruction, runtime epistemic gate, and oracle.

UCR is the fraction of retrieval-required opportunities that emit an unsupported factual commitment. Authorized Commitment Coverage is the fraction of non-abstention retrieval-required opportunities answered with the authorized value. We also report exact accuracy, retrieval precision/recall and rate, confident-hallucination catch rate, quadrant retrieval, runtime enforcement, generated tokens, calls, and wall time. Accuracy intervals and complete threshold sweeps are machine-readable.

Table 1: Untouched 80-example Phase A test. Acc.=exact accuracy, Ret.=retrieval rate, Cov.=Authorized Commitment Coverage.

Model	Condition	Acc.	Ret.	UCR	Cov.
Qwen	LLM only	.238	.000	1.000	.000
	FLARE-like	.625	.500	.675	.325
	Semantic	.613	.513	.250	.750
	Conf. OR semantic	.838	.763	.250	.750
	Runtime gate	.975	.763	.000	1.000
	Upfront RAG	.850	1.000	.250	.750
SmolLM2	LLM only	.463	.000	1.000	.000
	FLARE-like	.463	.000	1.000	.000
	Semantic	.575	.513	.775	.225
	Conf. OR semantic	.575	.513	.775	.225
	Runtime gate	.963	.513	.000	1.000
	Upfront RAG	.400	1.000	.775	.225
Gemma	LLM only	.463	.000	1.000	.000
	FLARE-like	.813	.900	.325	.675
	Semantic	.800	.513	.300	.700
	Conf. OR semantic	.825	.913	.300	.700
	Runtime gate	.775	.913	.000	1.000
	Upfront RAG	.825	1.000	.300	.700

6 Phase A Results

All runs use greedy decoding, seed 17, pinned revisions, PyTorch 2.13.0+XPU, and Intel integrated graphics. Each family produces 1,456 prediction rows and 1,456 traces. Table 1 reports the primary conditions.

Confidence and semantic risk are not equivalent. Qwen confidence retrieves both low-confidence quadrants, giving 50% recall and 50% false intervention; the semantic rule retrieves all 40 required cases and one of 40 controls. SmolLM2's fitted confidence retrieves nothing. Gemma confidence has 97.5% recall but 82.5% false activation. The same semantic policy has 100% recall and 2.5% false activation on all three because it is independent of checkpoint logits.

The runtime gate eliminates unsupported commitments in every family. It also shows why reliability and answer accuracy must be separated: Qwen and SmolLM2 improve sharply, whereas Gemma drops five accuracy points below upfront RAG. Enforcement prevents unsupported acceptance but cannot guarantee that the base answer or abstention policy maximizes exact-match utility.

7 Phase B: Context, Weights, or Runtime

The adapter dataset contains 160 balanced training rows and 40 development rows with disjoint entity namespaces and values. It excludes every held-out benchmark entity and answer. Targets contain either a four-field `retrieve` block or `NO_RETRIEVAL`; no target contains an answer value. The overlap audit has zero train/development, held-out entity, or held-out value collisions.

Rank-8 adapters train for two epochs on Qwen, SmolLM2, and Gemma. Final development losses are 0.00028, 0.0004, and 0.0001. This apparent fit does not transfer. Table 2 compares held-out policy selection. Interface success additionally requires the correct entity, attribute, date, and source. Prompt-token values are measured condition totals; the runtime semantic rows use the Phase A answer template and are not a matched estimate of policy-only prompt overhead.

All adapters collapse to `NO_RETRIEVAL`: zero recall, no false activation, and 50% balanced se-

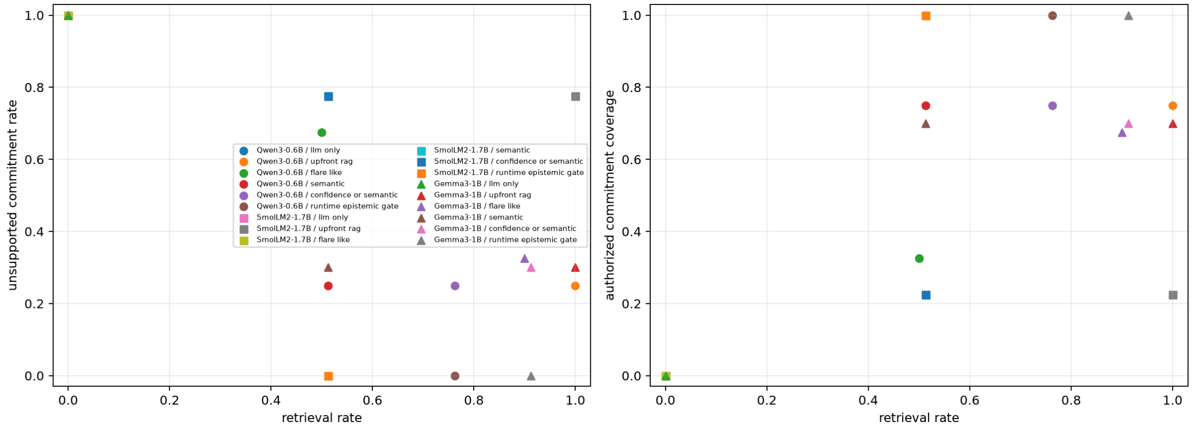

Figure 1: Retrieval cost against UCR and Authorized Commitment Coverage across three checkpoints. The runtime gate reaches zero UCR but has model-dependent accuracy consequences.

Table 2: Held-out 80-example placement evaluation. Sel.=selection accuracy, Int.=valid interface success, Rec.=retrieval recall, FA=false activation.

Model	Placement	Sel.	Int.	Rec.	FA	Prompt tok.	Gen. tok.
Qwen	Context	.500	.500	.000	.000	136.6	6.2
	Weights	.500	.500	.000	.000	67.6	6.0
	Runtime semantic	.988	.988	1.000	.025	101.4	10.2
SmolLM2	Context	.050	.013	.075	.975	172.2	31.4
	Weights	.500	.500	.000	.000	91.2	7.0
	Runtime semantic	.988	.988	1.000	.025	137.0	8.2
Gemma	Context	.863	.363	1.000	.275	139.0	24.4
	Weights	.500	.500	.000	.000	66.0	5.8
	Runtime semantic	.988	.988	1.000	.025	101.1	8.3

lection accuracy. Qwen context behaves the same. SmolLM2 context mostly over-triggers and often emits invalid blocks. Gemma context detects all needs but only 36.25% of all examples pass the complete interface criterion. Adding confidence to adapter outputs does not recover Qwen or SmolLM2 and inherits Gemma confidence's 82.5% false activation.

Near-zero development loss alongside zero held-out recall is direct evidence of protocol overfitting to narrow synthetic phrasing. It falsifies the proposed weight-placement benefit under this training design. More diverse training may change that result, but adapting after inspecting these held-out failures would require a new freeze.

8 Natural-Language Robustness Iteration

We next generate 192 balanced examples: 96 training, 32 development, and 64 untouched test cases. Eight retrieval-required categories cover release-specific facts, private identifiers, authoritative attribution, changed familiar facts, dynamic assignments, recorded fields, opaque fields, and post-reorganization assignments. Eight controls include supplied context, quoted freshness language, hypotheticals, electric current, historical-copy tasks, fictional sources, computation, and stable familiar facts. Train, development, and test use distinct base paraphrases. The freezer verifies zero exact duplicates, zero normalized template overlap, unique source entity–attribute keys, and consistent gold answers for split-specific controls.

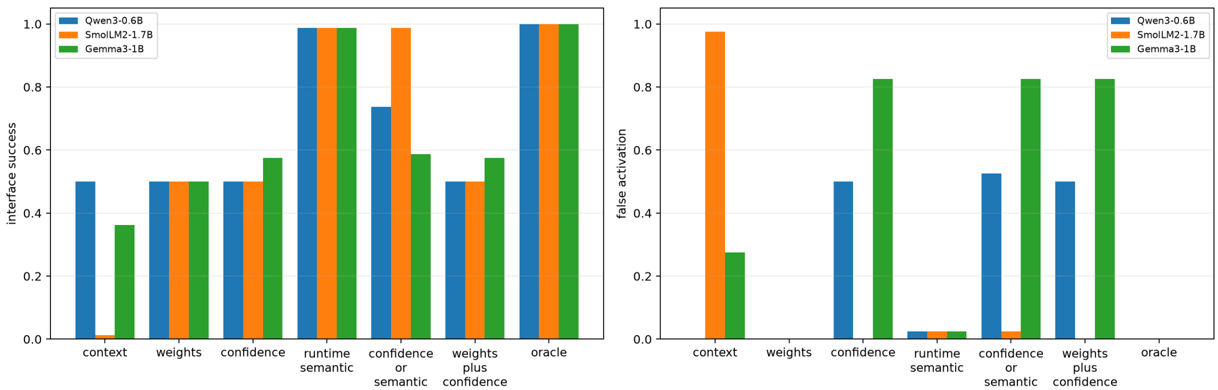

Figure 2: Interface success and false activation for context, weights, confidence, transparent runtime semantics, combinations, and oracle.

Four pre-freeze diagnostics are retained but excluded from claims. Version 1 failed the lexical screen at 93.75%. Version 2 passed at 87.5% but contained repeated questions. Version 3 removed duplicates but exposed source-key collisions between two categories. Version 4 fixed source keys but exposed split-template/gold-answer mismatches in several controls. Version 5 fixes these defects. Bag-of-words, word n-gram TF–IDF, and character n-gram TF–IDF logistic regressions score 81.25%, 85.94%, and 81.25% on its test set. This passes the predeclared 90% triviality screen, but 85.94% remains high and bounds the robustness claim.

8.1 Tokenization and lexical trigger strength

We compare the current word and character representations with a small Unicode tokenizer that splits words and punctuation. Separate conditions use the exact pinned tokenizer of each checkpoint. Token methods receive unigram and unigram–bigram TF–IDF logistic classifiers. We also build BM25 exemplar indexes with unigram and bigram terms; $k \in \{1, 3, 5, 10\}$ and the retrieval threshold are selected on development only. Raw and normalized native pieces are retained separately. No LLM call or recurring context token is used. Per-example pieces, predictions, vocabulary and index sizes, unknown-piece rates, latency, confusion matrices, and subclass scores are machine-readable.

This falsifies a broad reading of semantic-rule superiority on both freezes. All three pre-specified model-aligned n-gram conditions are perfect, including on historical-copy controls. Natural-v5 BM25 is also competitive: normalized Qwen and SmolLM2 pieces reach 1.000, but native-piece BM25 falls to .725 with .550 false activation on the original freeze. Thus native tokenization can expose highly discriminative surface structure, while the BM25 transfer result shows that model pieces are not uniformly better. Eighteen natural-v5 and 26 original-freeze conditions tie at perfect development accuracy, so an arbitrary development tie winner is not evidence of a unique production choice. The proper conclusion is benchmark lexical leakage/headroom, not general semantic content in BPE tokens. Runtime enforcement still determines what happens after a trigger and still yields zero UCR when recall is one.

Figure 3 shows that confidence calibration changes again: Qwen nearly reaches full recall but over-triggers, whereas SmolLM2 and Gemma miss required cases. The frozen semantic policy catches every required case but falsely retrieves four historical-copy controls containing both private IDs and reorganization language. We do not tune this error after inspecting test.

Evidence use is model-dependent. Advisory evidence yields authorized coverage of 81.25%,

Table 3: Matched trigger classification. Acc.=binary accuracy, Rec.=retrieval recall, FA=false activation. Native rows use raw model pieces and token unigrams+bigrams. The 80-example freeze uses its disjoint 160/40 policy train/development sets; natural-v5 uses its own 96/32 split.

		Original 80			Natural-v5		
Representation	Method	Acc.	Rec.	FA	Acc.	Rec.	FA
Current word	TF–IDF n-gram	.725	1.000	.550	.859	.969	.250
Current char	TF–IDF n-gram	.988	1.000	.025	.813	.875	.250
Shared NLP	TF–IDF n-gram	.725	1.000	.550	.875	1.000	.250
Qwen native	TF–IDF n-gram	1.000	1.000	.000	1.000	1.000	.000
SmolLM2 native	TF–IDF n-gram	1.000	1.000	.000	1.000	1.000	.000
Gemma native	TF–IDF n-gram	1.000	1.000	.000	1.000	1.000	.000
Semantic runtime	Fixed rules	.988	1.000	.025	.938	1.000	.125

Table 4: Natural-v5 test trigger and enforcement results. Rec.=retrieval recall, FA=false activation, Adv.=evidence advisory, Contract=explicit support contract.

Model	Conf. Rec.	Conf. FA	Sem. Rec.	Sem. FA	Adv. UCR	Contract UCR	Runtime UCR	Runtime Cov.
Qwen3-0.6B	.969	.625	1.000	.125	.188	.125	.000	1.000
SmolLM2-1.7B	.813	.000	1.000	.125	.125	.125	.000	1.000
Gemma3-1B	.781	.094	1.000	.125	.031	.000	.000	1.000

87.5%, and 96.88%. The explicit support contract lowers UCR but Qwen over-abstains, reducing its authorized coverage to 21.88%; SmolLM2 and Gemma retain 87.5% and 81.25%. Runtime enforcement copies a unique supported value or abstains otherwise, reaching zero UCR and full authorized coverage for all three. Its exact accuracy is 81.25%, 90.62%, and 68.75%, versus 37.5%, 46.88%, and 25% without retrieval and 87.5%, 96.88%, and 75% for the source-copy oracle. Retrospective verification detects all wrong required forecasts but, without a forced rewrite, leaves UCR at 100%. Detection is therefore not correction. Per-condition token, call, retrieval-time, and wall-time totals remain machine-readable in the released summaries and traces.

A deterministic four-document controller stress test has three factual opportunities per document: a control, a later emerging evidence need, and reuse of the same evidence. It records 100% early prospective catch, 100% late retrospective detection, 50% reuse across required opportunities, no repeated unnecessary retrieval, and zero runtime UCR. This tests interruption and cache mechanics, not free-running long-form generation; it is not evidence that models discover all factual opportunities in unconstrained prose.

The learned-policy gate remains NO-GO. Natural-v5 is now consumed, shallow lexical accuracy remains high, semantic rules leave only false-activation headroom, and the prior adapters failed held-out retrieval despite near-zero development loss. Reopening LoRA requires richer training data and a new test freeze created before adaptation.

9 Paper 2.5 Gate

The pre-registered gate asks whether at least two families reduce UCR with the runtime gate while retrieving less often than upfront RAG. Qwen, SmolLM2, and Gemma all pass; Paper 2.5 is therefore GO. This authorizes testing source heterogeneity. It does not authorize learned source routing, because Phase B provides no evidence that the retrieval policy should move into weights.

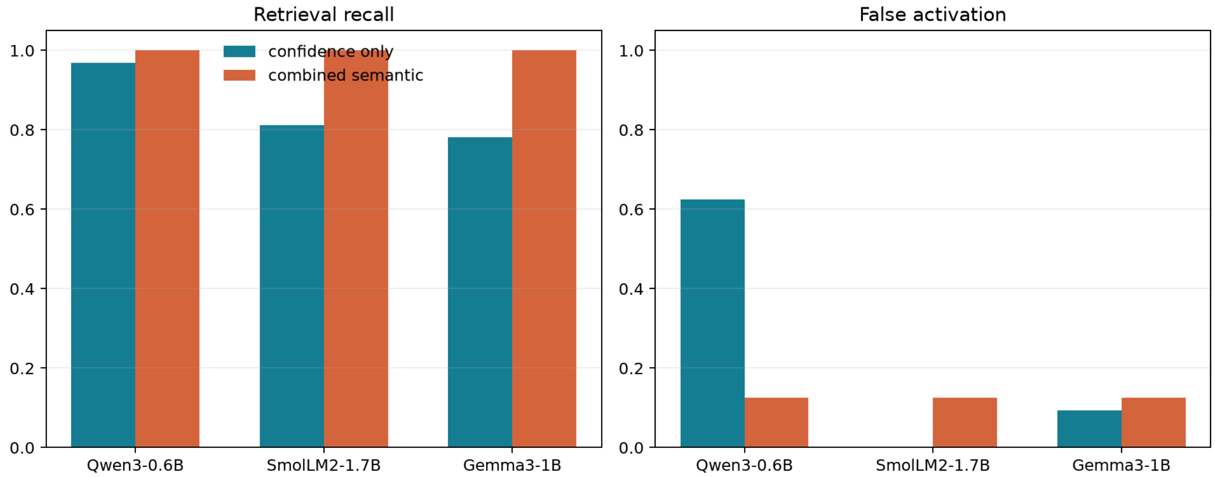

Figure 3: Checkpoint confidence versus the unchanged transparent semantic policy on the 64-example natural-v5 test.

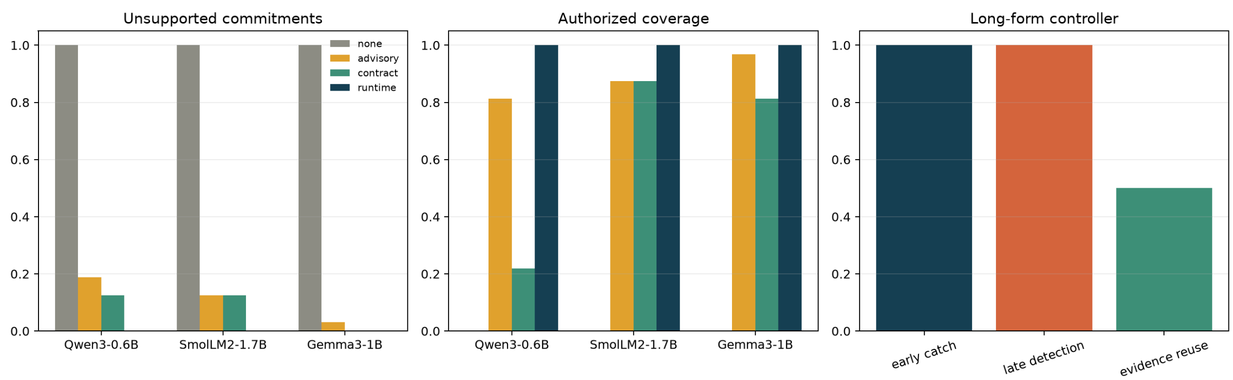

Figure 4: Natural-v5 enforcement ladder and deterministic multi-opportunity controller stress test. Runtime enforcement is reliable relative to the configured source; it is not a truth oracle.

10 Falsification and Limitations

The primary result would be weakened if a better-calibrated FLARE implementation matches semantic routing at equal cost, if natural-language semantic rules lose precision, or if the confident-risk quadrants disappear on new data. The present result instead finds checkpoint-dependent confidence behavior and stable rule recall, but only on one controlled source and mostly synthetic short answers.

The benchmark is selected using Qwen confidence, so its exact 20-per-quadrant balance does not transfer to other checkpoints. SmolLM2 lacks low-confidence test examples. There is one seed per model and no ranking noise. The natural extension passed its limited lexical screen only narrowly; the tokenizer-aware audit subsequently finds perfect model-token classifiers, so that screen was insufficient. Its long-form component is a deterministic controller simulation, not unconstrained generation. The semantic cue distribution remains controlled. The runtime gate is deterministic and answer-aware relative to source evidence, not a production truth system. Full-prefix generation latency is not production latency.

Phase B uses narrow train templates despite strict entity/value leakage controls; its failure may be covariate shift rather than a general limitation of LoRA. Conversely, development loss is clearly not evidence of interface generalization. The ICL baseline uses textual examples rather than native tool schemas. No condition trains or tests multi-source selection.

11 Reproduction

Paper-specific code is under `src/ccpu/paper1_5`. From the repository root, activate the validated XPU environment and run:

```
$py = 'C:\Users\j.simao\.venvs\modal-llm-xpu\Scripts\python.exe'
$env:PYTHONPATH = (Resolve-Path src).Path
& $py -m ccpu paper1.5 freeze-next \
  --config configs/paper1_5/next_iter_freeze_qwen_xpu.json \
  --output-dir artifacts/paper1_5/next_iter/freeze_qwen_v2
& $py -m ccpu paper1.5 run-hf \
  --config configs/paper1_5/next_iter_qwen_xpu.json \
  --output-dir artifacts/paper1_5/next_iter/qwen_v2
python -m ccpu paper1.5 analyze-next \
  --config configs/paper1_5/next_iter_analysis.json \
  --output-dir artifacts/paper1_5/next_iter/analysis_v1
& $py -m ccpu paper1.5 train-policy-lora <arguments in README>
python -m ccpu paper1.5 analyze-policy \
  --config configs/paper1_5/policy_placement.json \
  --output-dir artifacts/paper1_5/next_iter/policy_analysis_v1
python -m ccpu paper1.5 freeze-natural \
  --config configs/paper1_5/natural_robustness_xpu.json \
  --output-dir artifacts/paper1_5/natural_v5/data
& $py -m ccpu paper1.5 run-natural <natural-v5 paths and model label>
python -m ccpu paper1.5 analyze-natural \
  --config configs/paper1_5/natural_analysis.json \
  --output-dir artifacts/paper1_5/natural_v5/analysis
& $py -m ccpu paper1.5 compare-natural-tokenizers \
  --benchmark artifacts/paper1_5/natural_v5/data/benchmark.jsonl \
  --tokenizer-config configs/common/tokenizer_triggers.json \
  --output-dir artifacts/paper1_5/tokenizer_triggers/natural_v5
& $py -m ccpu paper1.5 compare-natural-tokenizers \
  --benchmark artifacts/paper1_5/next_iter/qwen_v2/benchmark.jsonl \
  --train artifacts/paper1_5/next_iter/policy_data_v2/train.jsonl \
  --dev artifacts/paper1_5/next_iter/policy_data_v2/dev.jsonl \
  --tokenizer-config configs/common/tokenizer_triggers.json \
  --output-dir artifacts/paper1_5/tokenizer_triggers/original80
```

Manifests hash configurations, benchmark/source snapshots, predictions, traces, summaries, and adapter files and record checkpoint revisions, device, package, repository, and environment provenance.

12 Conclusion

Semantic epistemic risk adds information beyond checkpoint uncertainty in this controlled one-source regime and retains full recall under a more varied natural-language freeze. It does not beat model-aligned token n-gram triggers: their perfect scores reveal that the present freezes remain lexically separable under stronger representations. Advisory and instructed evidence use remain checkpoint-dependent; deterministic runtime enforcement eliminates unsupported commitments for all three small families. The placement extension is negative: context is inconsistent and the tested adapters overfit. Exact source mechanics and enforcement belong in runtime under these freezes, and trigger selection should remain CPU-side and auditable until a richer independently frozen benchmark separates lexical cues from epistemic semantics.

References

[1] Z. Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023. `https://arxiv.org/abs/2305.06983`.